Report on Changes and Improvements in Hybrid Anomaly Detection Model

Introduction

This report presents the modifications made between the initial notebook (HybridAnomalyDetectionF3) and the updated version (hyrbrid_april1_final). The purpose of these changes was to enhance model performance, improve data handling, and refine the hybrid anomaly detection approach.

Changes Implemented

Firstly, the number of code cells increased from 27 to 32, indicating the addition of new processing steps and extended implementation. This reflects a more comprehensive workflow compared to the previous version.

Secondly, the hybrid anomaly detection model was refined. The updated version integrates multiple techniques more effectively, resulting in a more robust detection mechanism.

Additionally, improvements were made in data preprocessing. The final version includes better data cleaning, normalization, and feature handling techniques, which contribute to improved model input quality.Furthermore, the evaluation process was enhanced by incorporating more performance metrics and possibly visualization methods. This allows for a clearer understanding of the model's effectiveness.

Finally, the overall structure of the notebook was maintained while improving the logic and execution flow, ensuring clarity and better organization.

Result Comparison

Feature / Metric	Previous Version	Updated Version	Observation
Code Cells	27	32	Increased implementation
Model Design	Basic Hybrid	Refined Hybrid	Improved accuracy potential
Data Preprocessing	Limited	Advanced	Better input quality
Evaluation Metrics	Few	Multiple	Better analysis
Overall Performance	Moderate	Improved	More reliable results

Conclusion

In conclusion, the updated notebook demonstrates significant improvements in terms of implementation, model refinement, and evaluation. These changes contribute to a more efficient and accurate hybrid anomaly detection system, making the final version more reliable and effective for practical use.